# 1 One sample estimation of centre

Last week we learned about the tests for center of one sample, the non-parametric equivalent of t-test.

Now we look at estimation. The point estimate of the population median can be taken as the sample median.

The question is about interval estimate. We also know from general inference that hypothesis tests can be inverted to get confidence intervals.

Let $\theta$ be the median.

$H_0 : \theta = a$ then $Y = \#(X_i < a) \sim \text{Bin}(n, \theta)$. Do not reject $H_0$ if

$$\text{qbinom}(\alpha/2, n, 0.5) < Y < \text{qbinom}(1 - \alpha/2, n, 0.5)$$

. Inverting the test means: Find the set of $a$ for which the above statement is true.

Example: The 2008-09 nine-month academic salary for Assistant Professors, Associate Professors and Professors in a college in the U.S. The data were collected as part of the on-going effort of the college's administration to monitor salary differences between male and female faculty members.

```
library(carData)
data("Salaries")
qbinom(0.025, 397, 0.5) #179
qbinom(0.975, 397, 0.5)  #218
sort(Salaries$salary)[c(179,218)] #(105000, 111350)
```

What is the $a$ so that $k$ many $X_i$ are less than $a$? It is the $k$-th order statistic. So the confidence interval is $(X_{(179)}, X_{(218)})$

When $n$ is large, the qbinom quantities can be approximated by $n/2 \pm \Phi^{-1}(\alpha/2)\sqrt{n/4}$ as under $H_0$, E$(Y)$=$n/2$ and Var$(Y)$=$n/4$.

# 2 Wilcoxon Rank Sum Test

Now we start we testing for equality of centre of two samples, the non-parametric equivalent of the two-sample t-test when the parent distributions are not normal and the sample sizes are small.

Samples must be random and independent. Probability distributions must be continuous.

- Denote the distributions by $D_1$ and $D_2$.
- The one sided hypotheses are as follows.

$$H_0 : D_1 \text{ and } D_2 \text{ are identical.}$$
$$H_a : D_1 \text{ is shifted right of } D_2 \text{ or } H_a : D_1 \text{ is shifted left of } D_2$$

- The two sided hypotheses are as follows.

$$H_0 : D_1 \text{ and } D_2 \text{ are identical.}$$
$$H_a : D_1 \text{ is either shifted right or shifter to left of } D_2.$$

**Algorithm**

- We have two independent samples $X_1, \cdots, X_{n_1}$ and $Y_1, \cdots, Y_{n_2}$.
- Without loss of generatily let $n_2 < n_1$.
- Pool the sets together and rank them.
- Find the sum $(X)$ of ranks of the $X_i$'s in the pooled sample.
- The test statistic $T = X - n_1(n_1 + 1)/2$.

**Distribution of test statistic under the null hypothesis** The exact distribution, critical region and p-values are complicated and can be obtained through permutation methods.

- Reaction Times of Subjects Under the Influence of Drug A or B

X=(2.07, 2.11, 2.43, 2.50, 2.71, 2.84, 2.88)
Y=(1.62, 1.71, 1.93, 1.96, 2.24, 2.41)

- The ranks of $Y$ in the pooled sample are $5, 6, 9, 10, 11, 12, 13$.
- So $T = 38, m = 7, n = 13$.
- The p-value obtained from R is 0.01399.

**Permutation test**

There is no reason whatsoever why a permutation test has to use any particular test statistic. Any test statistic will do!

Consider, for example, the usual test statistic for a two-sample z test. Let us take the same example of Permeability constants and construct a two sample z statistic.

If the original data points are from a normal distribution then this follws a t-distribution. If the sample sizes are large, then irrespective of the parent distribution, this follows a normal distribution approximately.

However, if we cannot assume normality and the sample sizes are small (in our case 10 and 5), we can still generate the distribution of the statistic through all possible permutations of the data.

Thus we can use the idea of permutation to generate the null dstribution of any statistic.

What about the one sample z statistic?

## 3 Mann-Whitney U-statistic

The Wilcoxon rank sum test statistic can be written as

$$T = \sum_{i,j} I(X_i > Y_j)$$

Show this.

Such statistics are called U-statistics, U for unbiased. Each of the building blocks is a bernoulli with success probability $P(X_1 > Y_1)$. Under the null hypothesis, this is 1/2. There are $n_1 n_2$ terms in the sum. The expectation and variance of $T$ are $n_1 n_2/2$ and $n_1 n_2(n_1 + n_2 + 1)/12$ respectively.

**Large Sample Distribution** When the sample sizes $m$ and $n$ are both large, we can use the normal distribution for the standardized test statistic

$$Z = \frac{T - \frac{n_1 n_2}{2}}{\sqrt{\frac{n_1 n_2 (n_1 + n_2 + 1)}{12}}}$$